

Incidence and Impact of Housing Subsidies in Urban Slums: Evidence from India

Sabareesh Ramachandran

Indian School of Business

Targeting welfare benefits at poorest households is a challenge

Participation costs or ordeals can affect who takes up the benefit

- Participation costs screen out those who value the benefit the least [Nichols & Zeckhauser, 1982; Alatas et al., 2016]
- But costs like application difficulties can also exclude the poorest who cannot afford them [Deshpande & Li, 2019; Gupta, 2026]

Targeting welfare benefits at poorest households is a challenge

Participation costs or ordeals can affect who takes up the benefit

- Participation costs screen out those who value the benefit the least [Nichols & Zeckhauser, 1982; Alatas et al., 2016]
- But costs like application difficulties can also exclude the poorest who cannot afford them [Deshpande & Li, 2019; Gupta, 2026]

When the participation cost scales with income, take-up can be progressive

- Known when participation requires giving up a better option: workfare [Besley & Coate, 1992; Ravallion, 1991], public schooling/healthcare [Besley & Coate, 1991]
- This paper: The participation cost can also be an asset which is correlated with wealth: the house you must give up to get a housing transfer

Targeting welfare benefits at poorest households is a challenge

Participation costs or ordeals can affect who takes up the benefit

- Participation costs screen out those who value the benefit the least [Nichols & Zeckhauser, 1982; Alatas et al., 2016]
- But costs like application difficulties can also exclude the poorest who cannot afford them [Deshpande & Li, 2019; Gupta, 2026]

When the participation cost scales with income, take-up can be progressive

- Known when participation requires giving up a better option: workfare [Besley & Coate, 1992; Ravallion, 1991], public schooling/healthcare [Besley & Coate, 1991]
- This paper: The participation cost can also be an asset which is correlated with wealth: the house you must give up to get a housing transfer

Targeting a program well is also important to maximize impact of the transfer

This paper studies India's national housing program

Over a billion people live in slums around the world [United Nations]

- LMICs govts support the poor with public housing programs
- Typically large lumpy transfers to meet up-front costs

This paper studies India's national housing program

Over a billion people live in slums around the world [United Nations]

- LMICs govts support the poor with public housing programs
- Typically large lumpy transfers to meet up-front costs

Pradhan Mantri Awas Yojana (PMAY)

- Aims to provide the urban poor a formal house on their own land
- Targeted at 20 million households costing $\sim$1–2% of annual budget
- Beneficiaries receive Rs 2 lakh (USD 2500) to spend on construction
- Must demolish existing structure to lay a foundation and build a house with concrete roof

Study incidence, impact, and welfare parameters

Study implementation in slums in Odisha to answer:

Study incidence, impact, and welfare parameters

Study implementation in slums in Odisha to answer:

Incidence: Are poorer households more/less likely to benefit?

- Relocation hassles + credit constraints may be harder for the poor

Study incidence, impact, and welfare parameters

Study implementation in slums in Odisha to answer:

Incidence: Are poorer households more/less likely to benefit?

- Relocation hassles + credit constraints may be harder for the poor

Impact: What is the causal effect of PMAY + tenure security on house quality & outcomes?

- Eligibility: HHs need to have some tenure documents to receive the benefits
- State govt handed non-transferable land rights certificates
- Only households living on state govt land received certificates
- Exogenous variation within slums in who could receive tenure documents + housing grant

Study incidence, impact, and welfare parameters

Study implementation in slums in Odisha to answer:

Incidence: Are poorer households more/less likely to benefit?

- Relocation hassles + credit constraints may be harder for the poor

Impact: What is the causal effect of PMAY + tenure security on house quality & outcomes?

- Eligibility: HHs need to have some tenure documents to receive the benefits
- State govt handed non-transferable land rights certificates
- Only households living on state govt land received certificates
- Exogenous variation within slums in who could receive tenure documents + housing grant

Welfare: Given a fixed budget, what grant size is optimal?

Preview of results

Incidence: Take-up is progressive despite participation costs

- Grant receipt falls from 23% to 18% from the 5th to the 95th income percentile
- Richer households opt out: quality of existing house is an opportunity cost to participating

Preview of results

Incidence: Take-up is progressive despite participation costs

- Grant receipt falls from 23% to 18% from the 5th to the 95th income percentile
- Richer households opt out: quality of existing house is an opportunity cost to participating

Impact: Large improvement in house quality among eligible households

- First stage: 12pp higher grant receipt
- House quality index improves by 0.09sd; 10pp increase in concrete roofs
- Houses better protected from heat and rainfall: LATE of -1.9°C on temperature inside the house on hot days

Preview of results

Incidence: Take-up is progressive despite participation costs

- Grant receipt falls from 23% to 18% from the 5th to the 95th income percentile
- Richer households opt out: quality of existing house is an opportunity cost to participating

Impact: Large improvement in house quality among eligible households

- First stage: 12pp higher grant receipt
- House quality index improves by 0.09sd; 10pp increase in concrete roofs
- Houses better protected from heat and rainfall: LATE of -1.9°C on temperature inside the house on hot days

Welfare: Holding budget fixed, bigger grant reduces welfare transferred

- Framework: take-up is sufficient statistic for welfare transferred
- Increasing grant size gives more to fewer recipients and draws in people who value the grant less

This paper contributes to four literatures

1. Participation costs that scale with wealth screen out richer households
    - Participation costs can screen out those who value the benefit the least [Nichols & Zeckhauser, 1982; Alatas et al., 2016]
    - Costs can also be unaffordable for the poorest leading to their exclusion [Gupta, 2026; Romero & Singh, 2023; Deshpande & Li, 2019; Finkelstein & Notowidigdo, 2019]
    - When costs scale with income like forgone private schooling [Besley & Coate, 1991] or outside earnings [Besley & Coate, 1992] richer people opt out
    - I show that the participation cost can be an asset you own, correlated with wealth, and leads to richer households opting out

This paper contributes to four literatures

1. Participation costs that scale with wealth screen out richer households
2. Large *in situ* housing transfers improve houses for the poor
    - Light-touch transfers like titles, materials, and services yield modest improvements [Field, 2007; Galiani et al., 2017; McIntosh et al., 2018]
    - Relocation programs disrupt networks and see low take-up [Barnhardt et al., 2017; Franklin, 2020]
    - Even *in situ* redevelopments cause disruptions and suffer implementation failures [Philip, 2026]
    - I find a large *in situ* conditional cash transfer delivers big gains in house quality
    - The improved house serves as an adaptation to extreme weather

This paper contributes to four literatures

1. Participation costs that scale with wealth screen out richer households
2. Large *in situ* housing transfers improve houses for the poor
3. Take-up is a sufficient statistic for welfare transferred and can be used to make policy decisions
    - Sufficient statistics recover welfare without a full structural model [Chetty, 2009; Finkelstein & Hendren, 2020]
    - MVPF values policies through their welfare and fiscal effects, but needs each outcome priced [Hendren & Sprung-Keyser, 2020]
    - I show that when participation is voluntary, the take-up rate is a sufficient statistic for welfare transferred, analogous to ACR in trade [Arkolakis et al., 2012]
    - Theory can be combined with take-up elicitation experiments to make optimal policy decisions

This paper contributes to four literatures

1. Participation costs that scale with wealth screen out richer households
2. Large *in situ* housing transfers improve houses for the poor
3. Take-up is a sufficient statistic for welfare transferred and can be used to make policy decisions
4. First evaluation of India's flagship housing program at scale
   - PMAY: 1–2% of the federal budget, 20 million target households, yet no causal evidence on who benefits or by how much
   - Programs at scale can differ from small pilots [Muralidharan & Niehaus, 2017; Allcott, 2015], and government implementation often weakens effects [Bold et al., 2018; Banerjee et al., 2017; Vivalt, 2020]
   - I present the first causal evaluation of PMAY's incidence and impact
   - Evaluate with no researcher involvement in program delivery
   - Estimate medium-run effects, nine years after the program began

Context and Intervention

Slums are ubiquitous in urban India

One in six people in urban India live in slums

Slums develop organically with rural-urban migration

- Poor migrants occupy vacant lands in the city
- They start with tents and over time build a better structure

Slum houses are of poor quality

- Study house: 8% have concrete roof, 40% have no access to toilets
- Water-logging during rains and vulnerable to extreme weathers
- Residents susceptible to communicable and waterborne diseases
- Not conducive for household business or studying at home

Slum dwellers have no land documents

- They build de facto land rights by living there for long
- Gain access to services like electricity and water over time
- No legal land rights or tenure security
- Houses sometimes sold in informal markets

PMAY is India's flagship housing scheme

Eligibility

- Need to have a title or land rights document
- Should belong to Economically Weaker Section

Benefits

- HHs receive Rs 200,000 (USD 2500) to construct a house
- Can construct at any cost and design meeting basic requirements
- Median cost of construction: Rs 300,000

Implementation mechanics

- Money transferred in 4 installments to bank account
- Each installment after completion of foundation, walls, roof, & house
- HHs need to break down existing structure to lay foundation
- Need to make temporary stay arrangements elsewhere

GoO program to give tenure documents

Odisha government introduced JAGA Mission

- A slum upgradation program to make cities slum-free

First phase: Give tenure documents to slum dwellers

- One-time program for residents as of 2017
- Tenure for the same land that they occupied then

Tenure is non-transferable

- Cannot be sold or transferred to another person
- May be bequeathed to children
- May be mortgaged but banks do not accept it

Study design

Identification from variation in eligibility

Classify slum households into two types

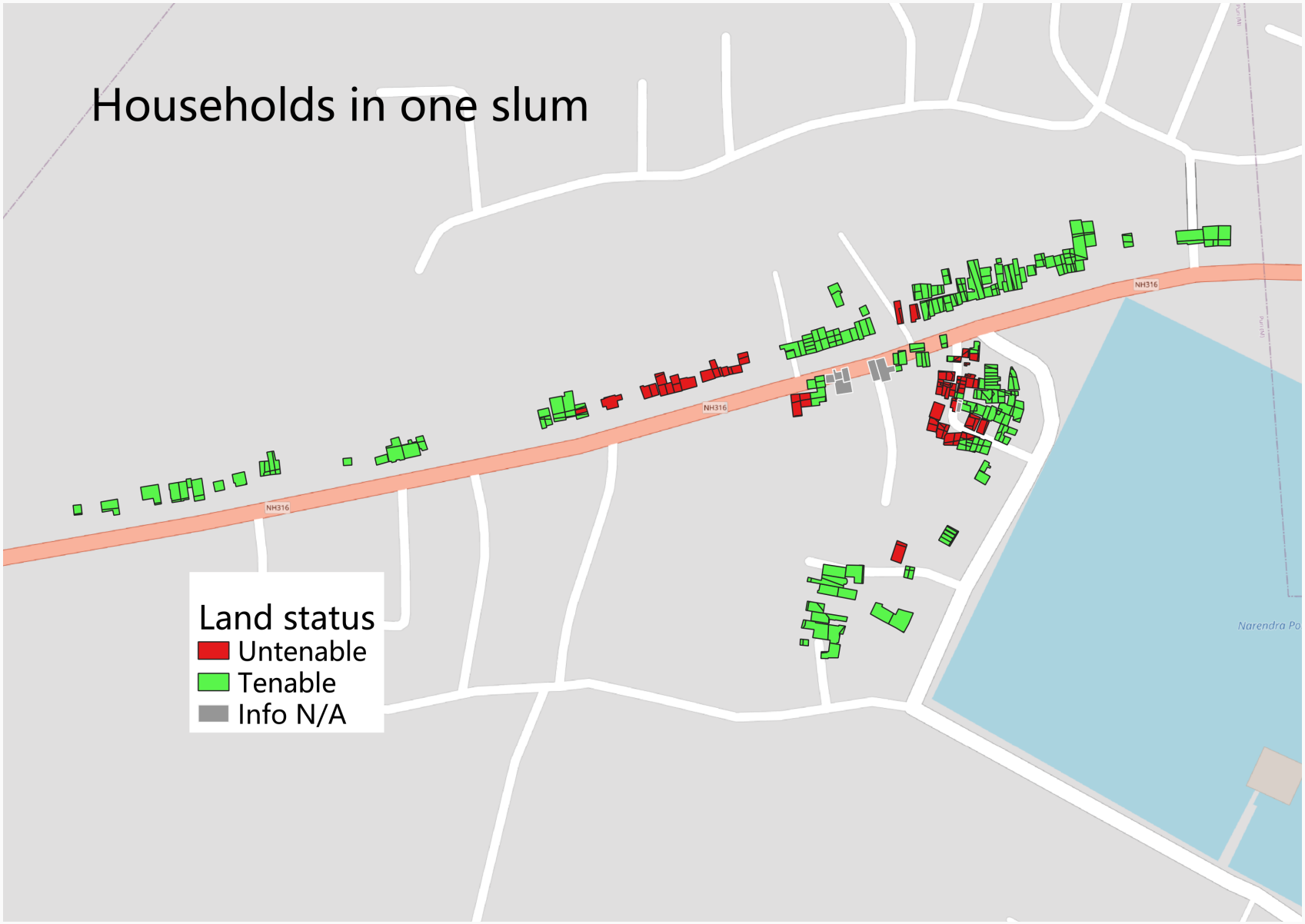
Tenable households:

- Living on lands owned by municipal govt or state govt
- Receive tenure documents immediately

Untenable households:

- Living on lands owned by private individuals, temples, royal families, railways, forest dept;
- Some cannot be given tenure documents
- A lengthy process of transferring ownership before documents can be given

Distribution of tenable and untenable HHs in one slum



Instrument is arguably exogenous

Strategy: Within the same slum compare households on tenable lands with households on untenable lands

Tenable status of land is exogenous to the HH conditional on slum

- Slum dwellers unlikely to know who owns the land at the time of settling down
- Eviction threats faced within the same slum are similar
- Slums on average are a few hundred meters wide-no change in location
- Land ownership decided many years ago before the slum dwellers occupied

$$Y_{is} = \beta\ tenable_{is} + \Gamma_s + \mu\ x_{is}^{BL} + \epsilon$$

Baseline characteristics not correlated with instrument

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Plot chars							
Occupied area (sqft)	1847	492.45	4154	506.57	20.265	-3.736 - 44.266	0.10
Area per person (sqft)	1847	186.16	4154	188.47	-0.758	-12.961 - 11.444	0.90
Income							
Annual income (Rs.)	1847	58813.47	4154	58494.21	-48.113	-3,605.878 - 3,509.651	0.98
Income per capita (Rs)	1847	20733.23	4154	21615.31	-604.246	-2,193.908 - 985.416	0.46
House quality							
Concrete roof	1847	0.06	4154	0.09	0.002	-0.018 - 0.022	0.85
Concrete/tiled roof	1847	0.30	4154	0.29	-0.009	-0.040 - 0.021	0.55
Brick wall	1847	0.52	4154	0.58	0.038	0.003 - 0.073	0.03*
Toilet in house	1847	0.32	4154	0.32	0.009	-0.023 - 0.041	0.57
Direct access to road	1847	0.76	4154	0.67	0.009	-0.019 - 0.037	0.54
Demographics							
HH size	1847	3.31	4154	3.33	0.095	-0.012 - 0.201	0.08
Number of children	1847	0.93	4154	0.95	0.080	-0.004 - 0.165	0.06
Average household age	1847	33.26	4154	33.12	-1.073	-2.161 - 0.015	0.05
Proportion female	1847	0.51	4154	0.50	-0.015	-0.034 - 0.003	0.10
Education							
Highest adult edu	1794	6.85	4035	7.33	0.224	-0.132 - 0.580	0.22
Mean adult edu	1797	5.08	4048	5.39	0.178	-0.117 - 0.473	0.24
Migration							
Years in slum	1759	59.57	3953	55.56	-0.286	-2.656 - 2.083	0.81

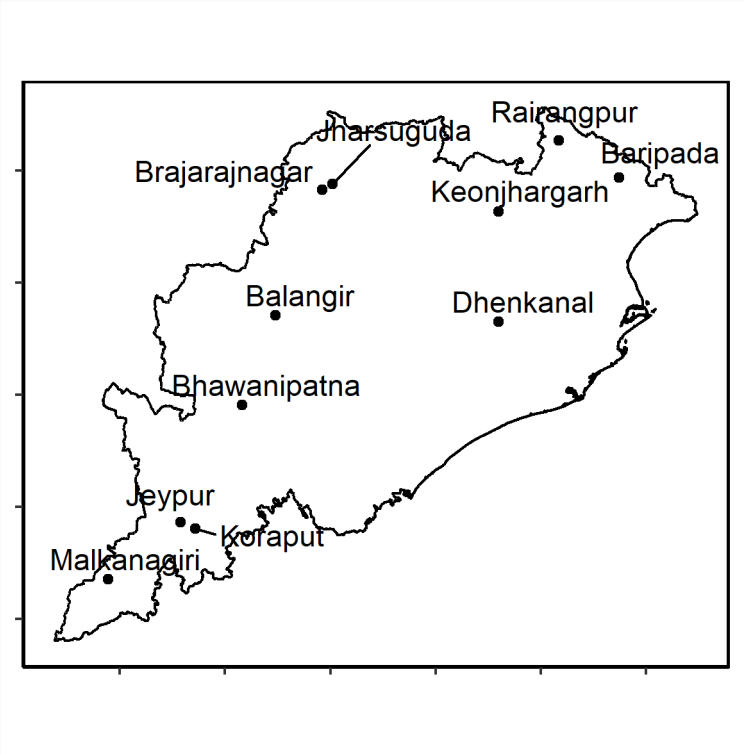
Round 1 sample | Full sample frame

Sample and data

Sample: 11 towns across Odisha, India

- Town population 50–200k
- Not the biggest cities

201 slums with 30 households each



Data description

Baseline: Govt census

- Conducted in 2017-18
- House quality, income, demographics

Instrument: Plot coordinates mapped to revenue data

Endline 1: Survey in Jan-Apr 2023

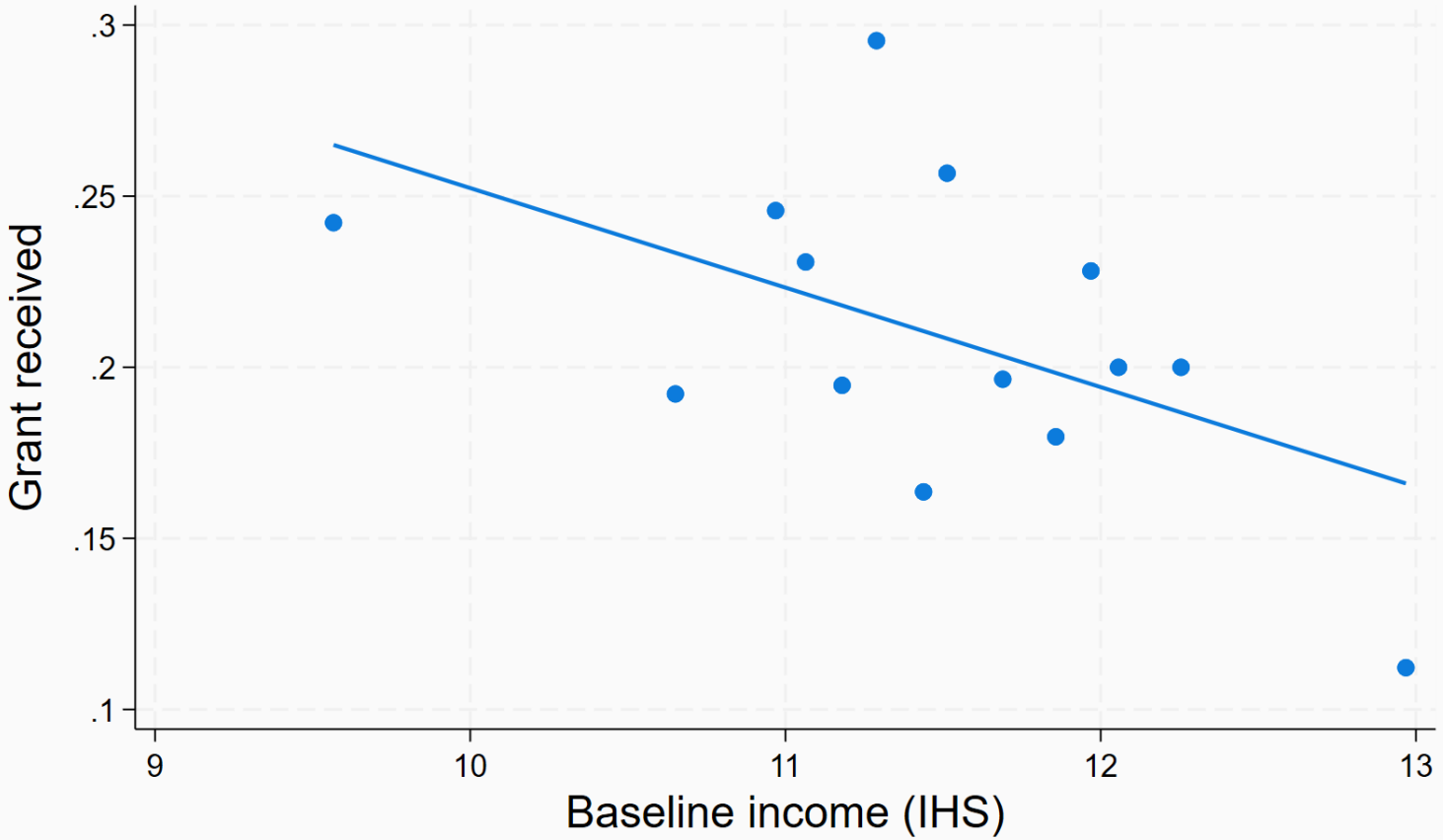
Endline 2: Survey in Jan-Apr 2026

- Take-up, house quality, outcomes

Results

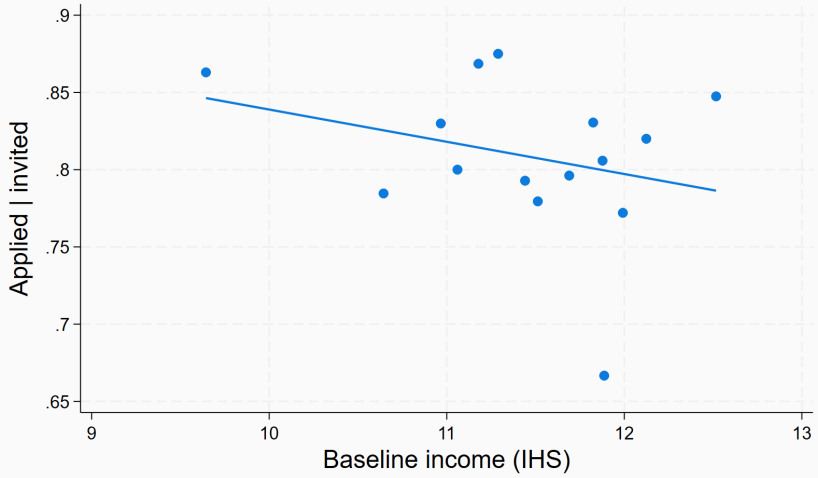
Incidence of the program is progressive

Share of households receiving the grant drops from 23% in the poorest quintile to 18% in the richest quintile

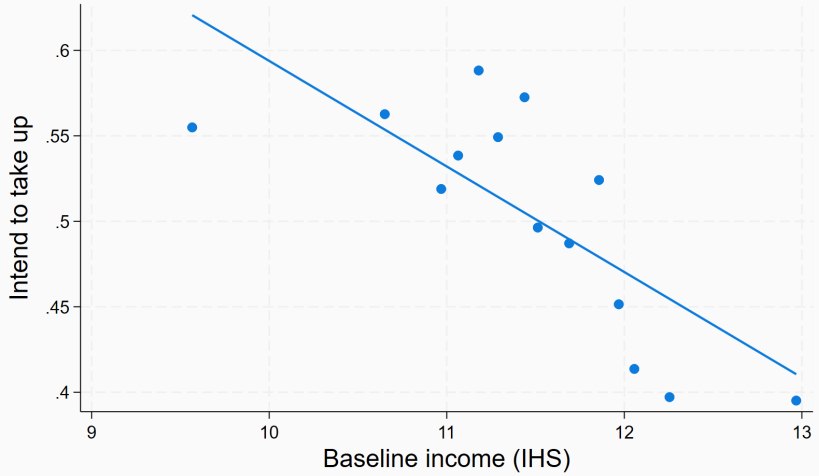


Richer households opt out of the program

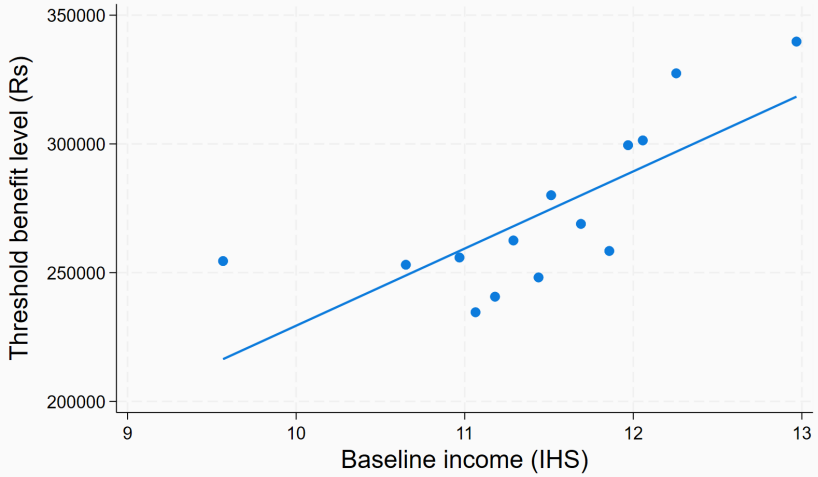
Applied | invited



Intend to take up



Threshold benefit



Poorer households more likely to

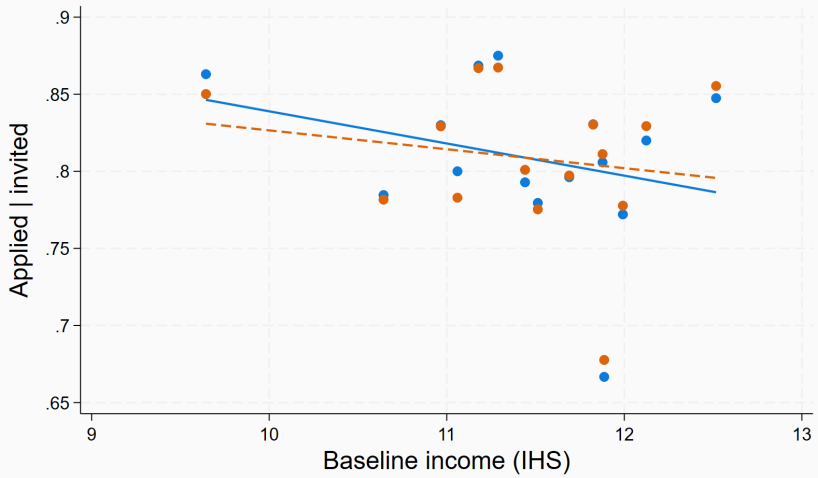
(i) apply when invited by govt officials,

(ii) report willingness to take up the program,

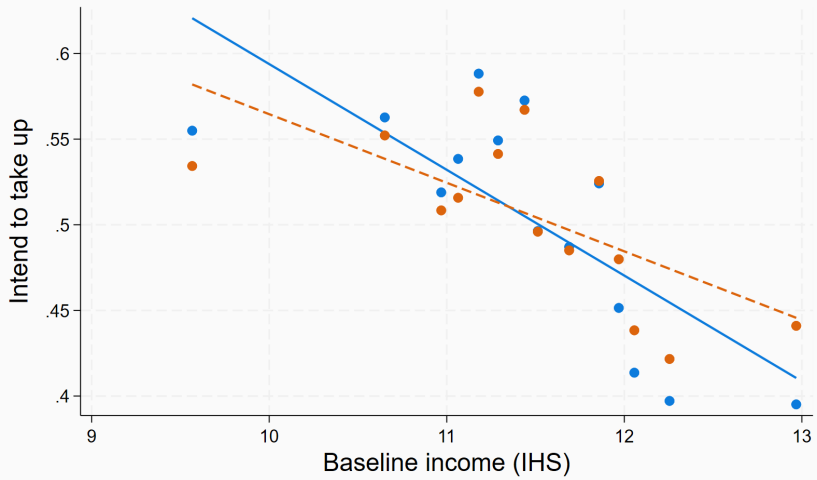
(iii) take up the program at lower benefit levels

Opt out partly mediated by BL house quality

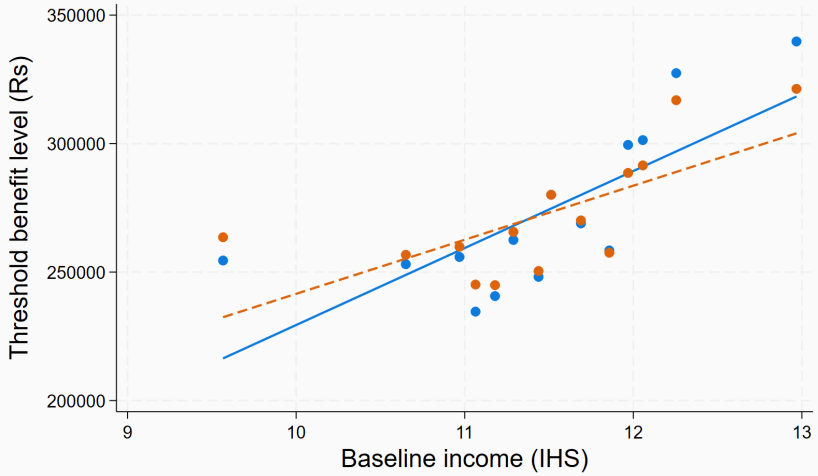
Applied | invited



Intend to take up



Threshold benefit



Raw

Net of baseline house quality

Progressivity flattens once we control for baseline house quality

Regression table | By income quintile

First stage: Instrument is relevant

	Received tenure document		PMAY		
	(1) Govt record (LRC)	(2) Self-report	(3) Received	(4) Money received	(5) Construction complete
Tenable	0.402***	0.143***	0.120***	18458.0***	0.0585***
	(0.0151)	(0.0140)	(0.0127)	(2197.3)	(0.00930)
Untenable mean	0.08	0.08	0.09	11711.96	0.04
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5962	6001	6001	5979	6001

Round 1

Significant ITT impact on house quality

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Tenable	0.0881*	0.0962***	0.0209	0.0274	0.00355	-0.000828
	(0.0352)	(0.0148)	(0.0157)	(0.0153)	(0.0151)	(0.0179)
Untenable mean	-0.00	0.18	0.76	0.76	0.66	0.44
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5744	6001	6001	6001	6001	6001

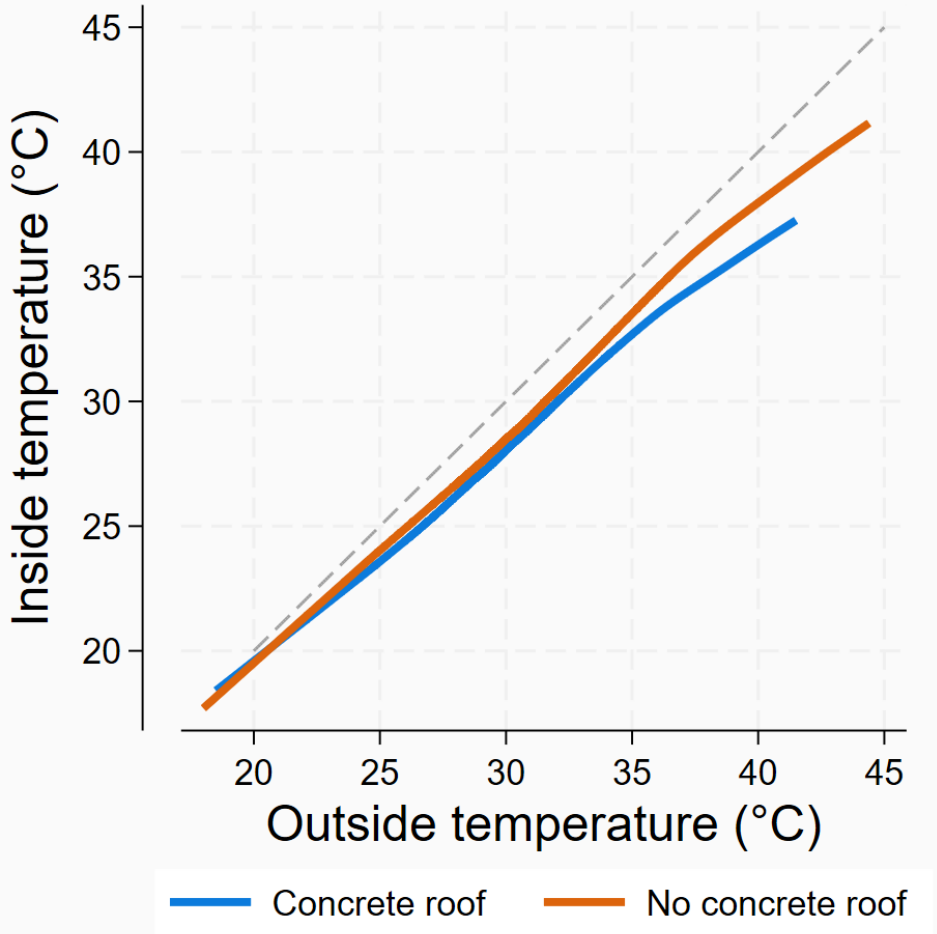
LATE | All outcomes | Index definition | Alternative indices

Improved house protects against extreme weather

	Satisfaction with house	Rainfall		Outside − inside temp	
	(1)	(2) No roof leak	(3) No waterlogging	(4) All days	(5) Hot days
Tenable	0.156**	0.0399*	-0.00528	0.0768	0.223†
	(0.0572)	(0.0174)	(0.0155)	(0.0545)	(0.118)
Untenable mean	3.84	0.23	0.80	1.53	1.59
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5987	5829	5829	5903	1851

LATE Round 1

Concrete roof provides protection against heat



No significant improvements in health and education

	Health		Education			Fertility	
	Fever	Diarrhoea	In school	Private	English	U5MR	Pregnancies
Tenable	0.00489	0.0378*	-0.0152	-0.0111	-0.00853	-0.0141	0.0349
	(0.0136)	(0.0148)	(0.0189)	(0.0168)	(0.0191)	(0.0304)	(0.0281)
Untenable mean	0.31	0.11	0.74	0.21	0.20	0.18	0.31
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	6000	3599	3762	3762	3128	1358	6001

No significant effects on economic outcomes

	Income	Any business	Consumption	Asset value	Outstanding loans
Tenable	-300.1	-0.00467	116.3	-1170.5	10188.7
	(811.1)	(0.0177)	(418.4)	(1188.0)	(8002.7)
Untenable mean	14167.80	0.29	11261.30	54325.93	36209.14
Slum FE	Yes	Yes	Yes	Yes	Yes
N	6001	6000	5919	5993	6001

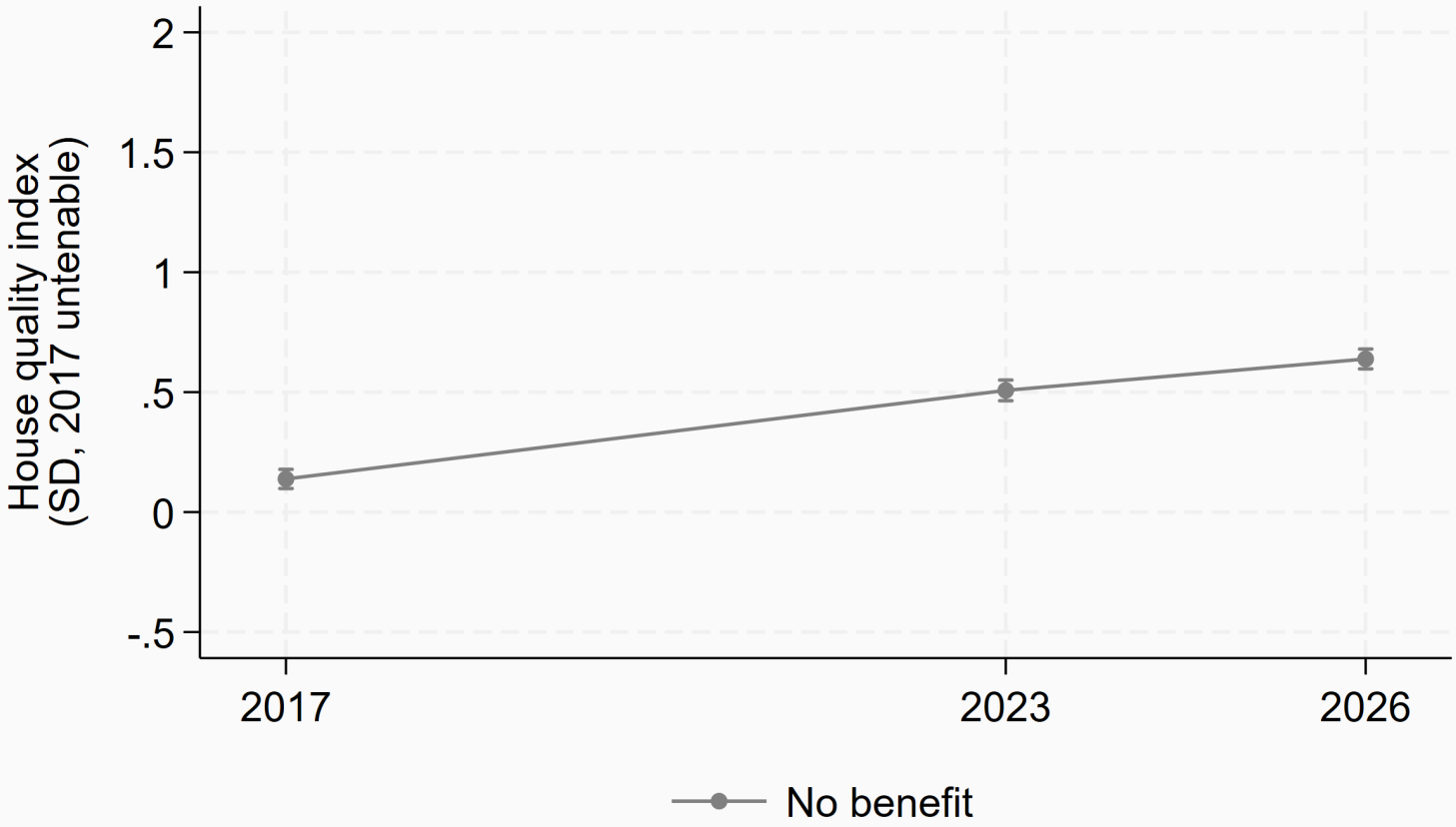
Direct effects of tenure security

Tenure security can directly impact house quality through:

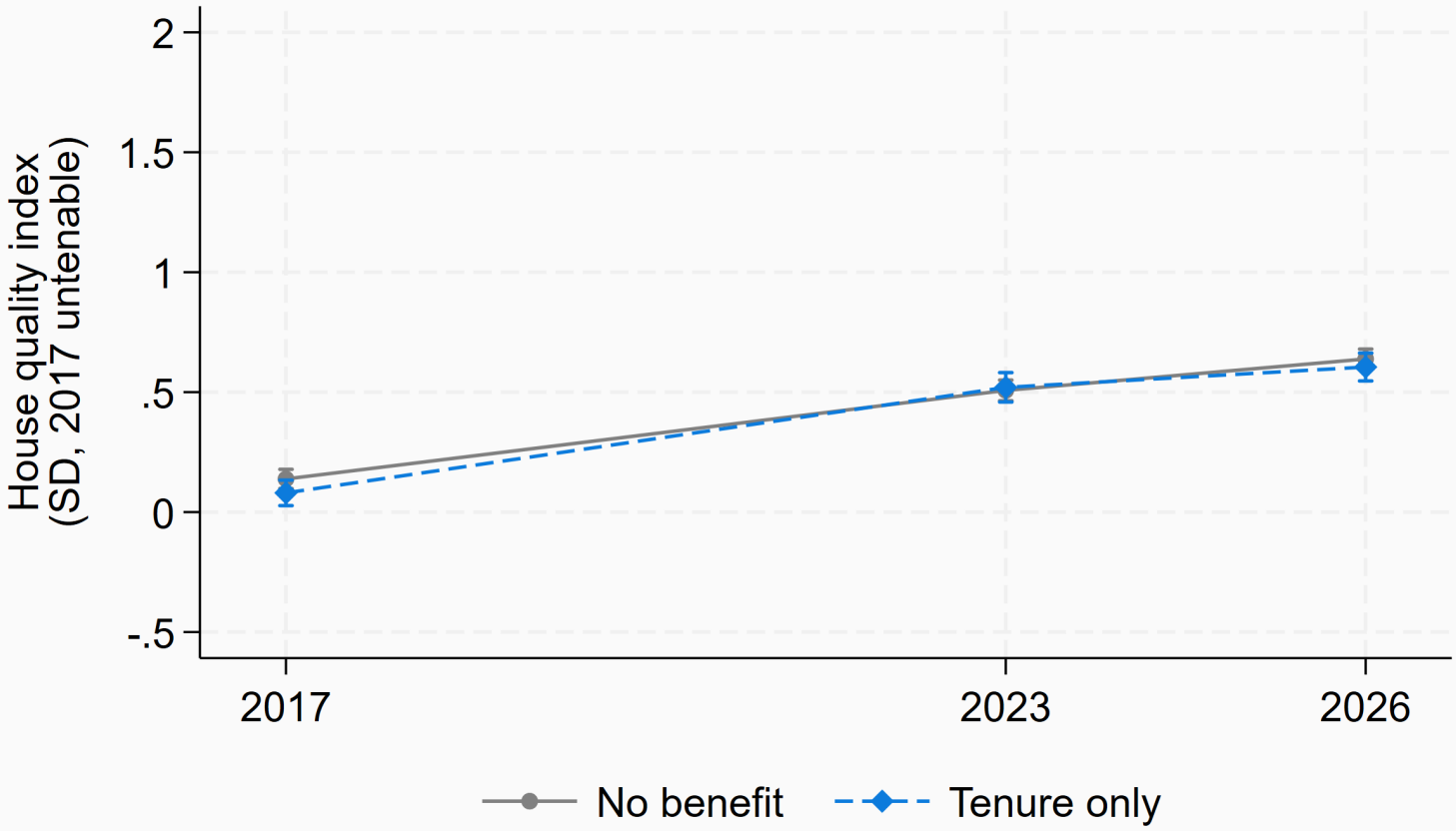
- Lower risk: reduced threat of eviction
- Access to credit: Mortgage land documents
- Wealth effect: Perceived ability to sell the house

	Received tenure doc	Have some tenure doc	Threat of eviction	Tried to mortgage	Able to mortgage	Able to sell land
	(1)	(2)	(3)	(4)	(5)	(6)
Tenable	0.136***	0.120***	-0.0547**	0.00147	0.00283	-0.00658
	(0.0144)	(0.0168)	(0.0174)	(0.00270)	(0.00184)	(0.00528)
Untenable mean	0.09	0.17	0.34	0.00	0.00	0.02
Tenable mean	0.29	0.40	0.29	0.01	0.00	0.01
Slum FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5824	5824	5850	5894	5894	5894

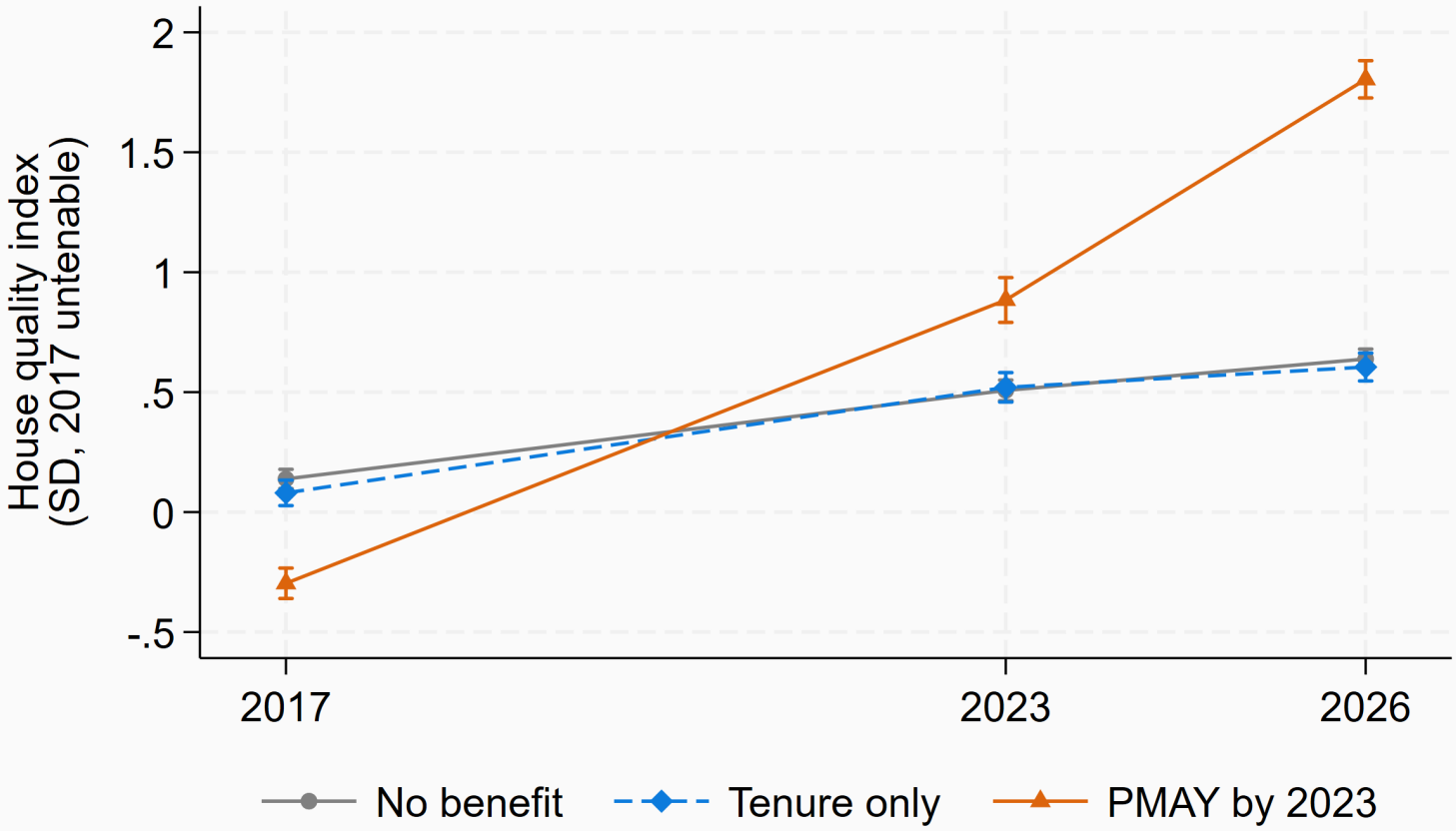
House quality rose when the grant arrived, not when tenure did



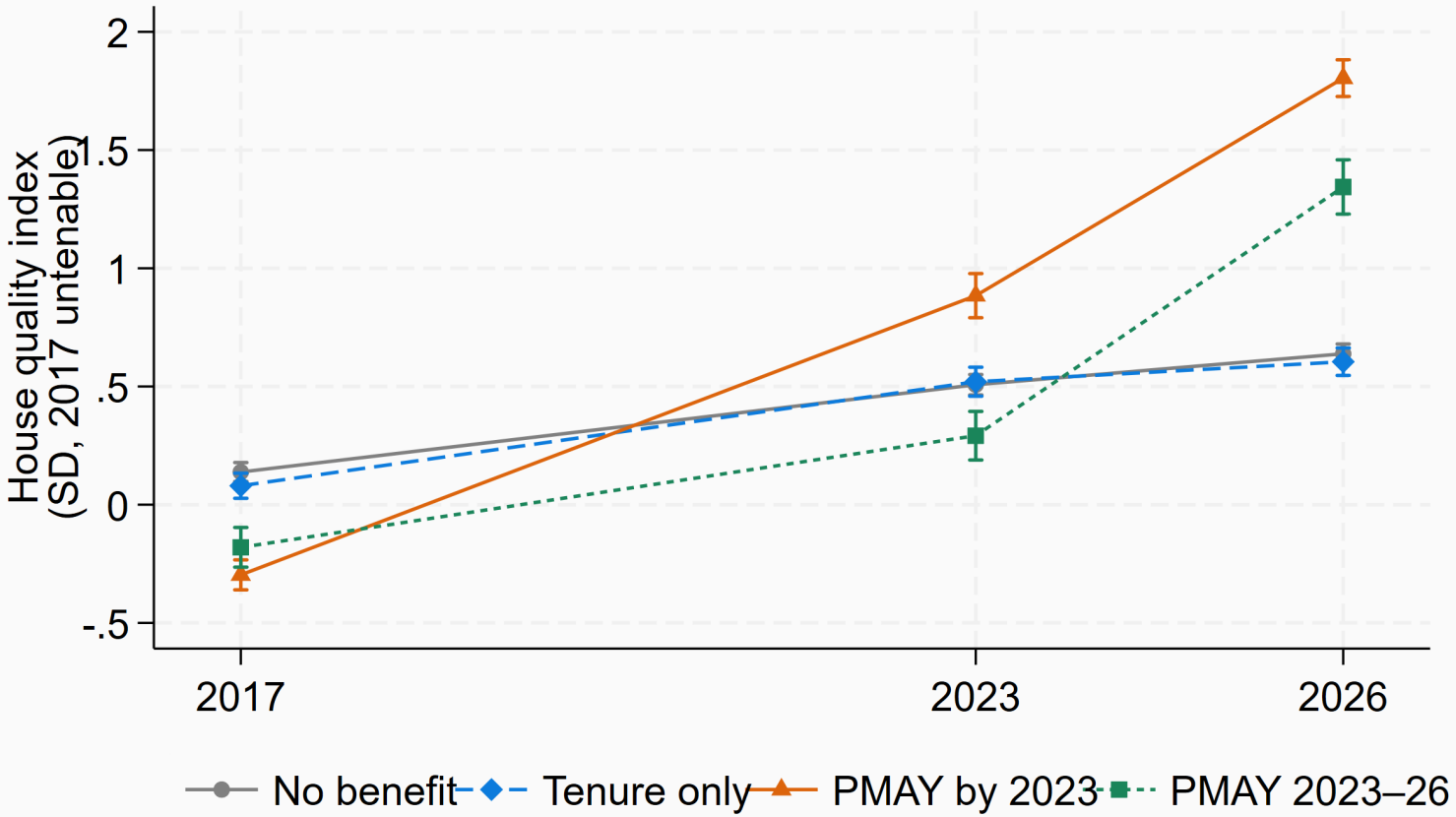
House quality rose when the grant arrived, not when tenure did



House quality rose when the grant arrived, not when tenure did



House quality rose when the grant arrived, not when tenure did



Welfare framework

Use take-up as revealed welfare gains

Policy question: Given a fixed budget, what maximizes welfare: small grant to more households or large grant to fewer households?

Use take-up as revealed welfare gains

Policy question: Given a fixed budget, what maximizes welfare: small grant to more households or large grant to fewer households? Calculating welfare benefits is hard

- Long time horizon of benefits
- Valuing costs of relocation and debt is hard

Use take-up as revealed welfare gains

Policy question: Given a fixed budget, what maximizes welfare: small grant to more households or large grant to fewer households? Calculating welfare benefits is hard

- Long time horizon of benefits
- Valuing costs of relocation and debt is hard

Use households' own decision to take up the grant

- A household takes up only if the benefit exceeds what it gives up
- Take-up rates reveal how much the program is worth to households

Notation

Household i with baseline income I_i and house quality h_i, grant size b

- Take up: utility $u_{yi} + \lambda_i$
- Do not take up: utility u_{ni}
- u_{yi}, u_{ni}: maximized utilities given baseline observables
- λ_i: idiosyncratic preference shock

Notation

Household i with baseline income I_i and house quality h_i, grant size b

- Take up: utility $u_{yi} + \lambda_i$
- Do not take up: utility u_{ni}
- u_{yi}, u_{ni}: maximized utilities given baseline observables
- λ_i: idiosyncratic preference shock

Take up if $u_{yi} + \lambda_i > u_{ni}$, so the welfare gain from the option to take up is

$$W_i = \max\left(u_{yi} + \lambda_i - u_{ni},\ 0\right)$$

Zero for non-takers, the net gain for takers

Assume distribution of preferences

Assumption: $\lambda_i \sim \text{Logistic}(0, \beta)$

- Equivalent to independent Gumbel shocks on u_{yi} and u_{ni}
- Natural for maximized value functions: each aggregates many choices with their own shocks
- Standard in discrete choice, trade, and migration [McFadden, 1974; Eaton & Kortum, 2002; Bryan & Morten, 2019]

Assume distribution of preferences

Assumption: $\lambda_i \sim \text{Logistic}(0, \beta)$

- Equivalent to independent Gumbel shocks on u_{yi} and u_{ni}
- Natural for maximized value functions: each aggregates many choices with their own shocks
- Standard in discrete choice, trade, and migration [McFadden, 1974; Eaton & Kortum, 2002; Bryan & Morten, 2019]

β measures preference heterogeneity

- Higher β: households with the same baseline observables differ more in what the grant is worth to them

Take-up rate is a sufficient statistic for welfare transferred

Take-up probability is logit:

$$p_i = \frac{1}{1 + e^{(u_{ni} - u_{yi})/\beta}} \quad \Longrightarrow \quad u_{yi} - u_{ni} = \beta \log\left(\frac{p_i}{1 - p_i}\right)$$

Take-up rate is a sufficient statistic for welfare transferred

Take-up probability is logit:

$$p_i = \frac{1}{1 + e^{(u_{ni} - u_{yi})/\beta}} \quad \Longrightarrow \quad u_{yi} - u_{ni} = \beta \log\left(\frac{p_i}{1 - p_i}\right)$$

Proposition: expected welfare transferred to household i is

$$E[W_i] = \beta \log\left(\frac{1}{1 - p_i}\right)$$

Take-up rate is a sufficient statistic for welfare transferred

Take-up probability is logit:

$$p_i = \frac{1}{1 + e^{(u_{ni} - u_{yi})/\beta}} \quad \Longrightarrow \quad u_{yi} - u_{ni} = \beta \log\left(\frac{p_i}{1 - p_i}\right)$$

Proposition: expected welfare transferred to household i is

$$E[W_i] = \beta \log\left(\frac{1}{1 - p_i}\right)$$

- (p_i, β) is sufficient: no need to specify the utility function
- Analogous to ACR in trade [Arkolakis et al., 2012]: take-up rate $\leftrightarrow$ domestic expenditure share, $\beta \leftrightarrow$ inverse trade elasticity

Intuition: high take-up means a high average gain

- If the program is worth a lot, nearly everyone takes up
- If the gain is close to zero on average, few take up
- The distribution of λ_i converts the take-up rate into a welfare value

Intuition: high take-up means a high average gain

- If the program is worth a lot, nearly everyone takes up
- If the gain is close to zero on average, few take up
- The distribution of λ_i converts the take-up rate into a welfare value

Most useful when many households decline the program

- As $p_i \to 1$, welfare diverges and becomes very sensitive to take-up
- In PMAY, about half of households say they would not take it up at the current grant

Intuition: high take-up means a high average gain

- If the program is worth a lot, nearly everyone takes up
- If the gain is close to zero on average, few take up
- The distribution of λ_i converts the take-up rate into a welfare value

Most useful when many households decline the program

- As $p_i \to 1$, welfare diverges and becomes very sensitive to take-up
- In PMAY, about half of households say they would not take it up at the current grant

Different distributional assumptions change the exact mapping, not the qualitative result

With a fixed budget, a larger grant has three effects on welfare

Budget B, grant b: at most B/b recipients

- If more apply ($n_a > B/b$), recipients are chosen by lottery

$$W = \frac{B}{b\, n_a} \sum_{i=1}^{n} \beta \log\left(\frac{1}{1 - p_i(b)}\right)$$

With a fixed budget, a larger grant has three effects on welfare

Budget B, grant b: at most B/b recipients

- If more apply ($n_a > B/b$), recipients are chosen by lottery

$$W = \frac{B}{b\, n_a} \sum_{i=1}^{n} \beta \log\left(\frac{1}{1 - p_i(b)}\right)$$

Raising b:

(+) Each recipient gets a larger transfer

(−) Fewer recipients

(−) Households with better houses, and smaller gains, select in

With a fixed budget, a larger grant has three effects on welfare

Budget B, grant b: at most B/b recipients

- If more apply ($n_a > B/b$), recipients are chosen by lottery

$$W = \frac{B}{b\, n_a} \sum_{i=1}^{n} \beta \log\left(\frac{1}{1 - p_i(b)}\right)$$

Raising b:

(+) Each recipient gets a larger transfer

(−) Fewer recipients

(−) Households with better houses, and smaller gains, select in

Net effect is ambiguous: it depends on who selects in at each grant size

Take-up rate elicited using price experiment

Each household asked whether it would take up the program at each grant size

- Everything else about the program held fixed
- Responses consistent: every household switches once, from no to yes
- Thresholds rise with income and house quality

Take-up rate elicited using price experiment

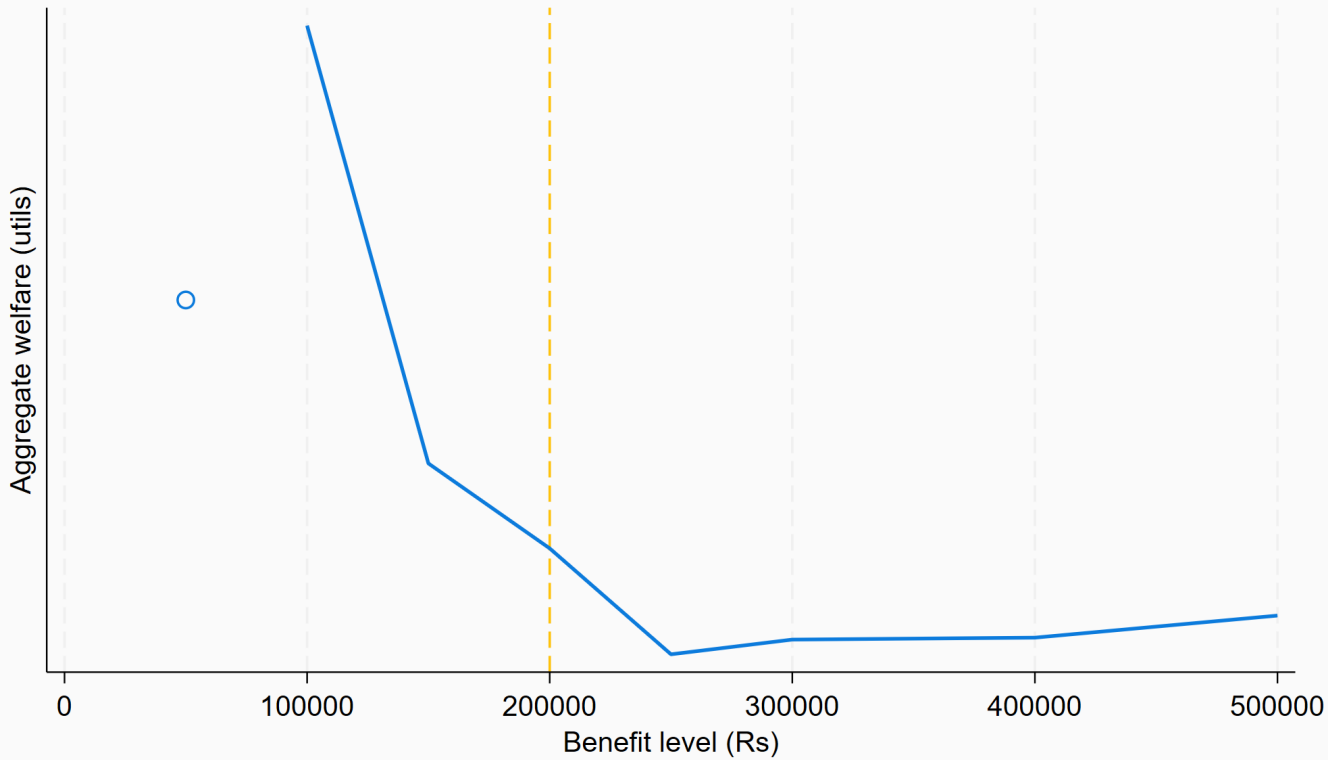
Each household asked whether it would take up the program at each grant size

- Everything else about the program held fixed
- Responses consistent: every household switches once, from no to yes
- Thresholds rise with income and house quality

Estimating p_i at each grant size

- 25 cells: baseline income quintile $\times$ baseline house-quality quintile
- About 200–300 households per cell
- p_i = share in the cell who would take up at grant size b

A smaller grant would transfer more welfare



Dashed line: current grant (Rs 200,000)

- Above the current grant, better-housed households select in and the number of recipients falls
- Lower grant: more recipients, and better screening toward those with the most to gain

▶ Normal shocks

Summary

When participation requires giving up an asset, richer households opt out

- PMAY is progressive despite requiring demolition and co-investment

The well-targeted program delivers large housing improvements

- Large gains in house quality, especially quality of roof
- Better protection from rain and heat: helps adapt to extreme weather

The grant size affects who takes up the program

- Take-up is a sufficient statistic for welfare transferred
- At the current budget, raising the grant would transfer *less* welfare: fewer recipients, weaker screening

Appendix

Balance: Full sample frame

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Occupied area (sqft)	16531	508.93	19317	509.58	-6.016	-15.898 - 3.865	0.23
Area per person (sqft)	12811	196.03	15581	198.36	-0.723	-6.987 - 5.540	0.82
Annual Income (winsorized)	14380	60618.89	17136	59895.58	789.965	-654.940 - 2,234.869	0.28
Income per capita (Rs)	12798	23164.28	15581	23168.80	537.454	-780.461 - 1,855.369	0.42
Permanent roof	14416	0.11	17122	0.10	0.005	-0.003 - 0.013	0.23
Brick wall	14422	0.60	17160	0.63	0.001	-0.012 - 0.013	0.92
Toilet in house	14421	0.42	17158	0.34	0.000	-0.012 - 0.012	0.97
Direct access to road	14421	0.75	17160	0.69	-0.006	-0.017 - 0.004	0.23
HH size	12823	3.24	15588	3.29	0.009	-0.037 - 0.055	0.70
Number of children	12816	0.91	15625	0.91	-0.028	-0.063 - 0.007	0.11
Average household age	12799	33.86	15585	33.63	0.015	-0.439 - 0.469	0.95
Proportion female	12822	0.51	15588	0.51	-0.008	-0.016 - -0.001	0.03*

All households in the 11 cities from the sampling frame.

◂ Back

Balance: Round-1 survey sample

	Untenable		Tenable				
	n	mean	n	mean	Diff	Confidence interval	p-value
Plot chars							
Occupied area (sqft)	1814	497.13	4103	507.41	16.989	-7.624 - 41.602	0.18
Area per person (sqft)	1814	189.71	4103	190.86	-1.661	-14.334 - 11.011	0.80
Income							
Annual income (Rs.)	1814	58878.10	4103	58852.27	-73.788	-3,787.355 - 3,639.780	0.97
Income per capita (Rs)	1814	20850.15	4103	21840.96	-651.480	-2,295.642 - 992.683	0.44
House quality							
Concrete roof	1814	0.06	4103	0.10	-0.003	-0.024 - 0.017	0.75
Concrete/tiled roof	1814	0.30	4103	0.29	-0.002	-0.033 - 0.029	0.90
Brick wall	1814	0.53	4103	0.58	0.014	-0.022 - 0.049	0.45
Toilet in house	1814	0.33	4103	0.33	0.008	-0.025 - 0.040	0.64
Direct access to road	1814	0.75	4103	0.68	0.022	-0.006 - 0.050	0.12
Demographics							
HH size	1814	3.29	4103	3.31	0.108	-0.001 - 0.217	0.05
Number of children	1814	0.93	4103	0.94	0.088	0.002 - 0.174	0.04*
Average household age	1814	33.39	4103	33.38	-0.971	-2.088 - 0.145	0.09
Proportion female	1814	0.51	4103	0.50	-0.017	-0.036 - 0.002	0.08
Education							
Highest adult edu	1704	6.68	3877	6.91	0.083	-0.298 - 0.463	0.67
Mean adult edu	1692	4.95	3858	5.16	0.113	-0.204 - 0.430	0.48
Migration							
Years in slum	1798	58.91	4066	54.19	0.797	-1.466 - 3.060	0.49

All households surveyed in the round-1 endline. Baseline covariates from the pre-intervention survey.

◂ Back

PMAY incidence by baseline income

	Grant received	Applied | selected		Intend to take up		Threshold benefit (Rs)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Baseline income (IHS)	-0.023**	-0.022	-0.016	-0.050***	-0.036***	22168***	16659***
	(0.007)	(0.012)	(0.012)	(0.009)	(0.009)	(3767)	(3802)
BL house-quality controls	No	No	Yes	No	Yes	No	Yes
Mean of outcome	0.210	0.818	0.818	0.511	0.511	269238	269238
N	6001	2369	2369	6001	6001	6001	6001

Notes. This table shows how the incidence of PMAY varies with baseline income. The dependent variable is whether the household had received the program when surveyed in round 2 in 2026 (column 1); whether it applied for the program conditional on being encouraged to do so by a local government official (columns 2–3); whether it stated it would like to take up the program if offered (columns 4–5); and the lowest benefit level at which it would be willing to take up the program (columns 6–7). The independent variable is the inverse hyperbolic sine of household income before the program; it is 10.3 at the 5th percentile and 12.3 at the 95th percentile. Six measures of the roof, walls and toilet before the program are added as controls in columns 3, 5 and 7. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

◂ Back

Selection into the program by baseline income quintile

Panel A: Short-run effects (round 1)

	Grant received	Applied | selected		Intend to take up		Threshold benefit (Rs)	
Income quintile 2	-0.0174	-0.0158	-0.0141	0.0118	0.0205	-682.1	-4120.7
	(0.0144)	(0.0273)	(0.0273)	(0.0187)	(0.0185)	(6967.2)	(6858.2)
Income quintile 3	-0.0380***	-0.0729***	-0.0698***	0.0130	0.0258	-3787.4	-9304.2
	(0.0131)	(0.0248)	(0.0249)	(0.0170)	(0.0169)	(6344.5)	(6258.0)
Income quintile 4	-0.0743***	-0.0408	-0.0277	0.00294	0.0204	1469.4	-6276.8
	(0.0161)	(0.0319)	(0.0318)	(0.0208)	(0.0206)	(7766.5)	(7649.5)
Income quintile 5	-0.0576***	-0.0488	-0.0372	-0.0468**	0.00468	35170.1***	12568.0*
	(0.0152)	(0.0310)	(0.0312)	(0.0196)	(0.0199)	(7325.3)	(7375.6)
BL house-quality controls	No	No	Yes	No	Yes	No	Yes
N	5917	2113	2113	5917	5917	5917	5917

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Panel B: Medium-run effects (round 2)

	Grant received	Applied | selected		Intend to take up		Threshold benefit (Rs)	
Income quintile 2	0.0103	0.0359	0.0384	0.0255	0.0284	-975.6	-1891.3
	(0.0167)	(0.0251)	(0.0251)	(0.0204)	(0.0203)	(8583.2)	(8539.3)
Income quintile 3	-0.0247	-0.0373	-0.0324	-0.0434*	-0.0331	12755.2	8975.6
	(0.0151)	(0.0227)	(0.0226)	(0.0185)	(0.0185)	(7801.6)	(7777.0)
Income quintile 4	-0.0489**	-0.0128	-0.00460	-0.0187	-0.00549	3804.4	-1626.6
	(0.0185)	(0.0278)	(0.0277)	(0.0227)	(0.0226)	(9553.4)	(9510.5)
Income quintile 5	-0.0410*	-0.0181	-0.00591	-0.131***	-0.0934***	63675.8***	48998.5***
	(0.0175)	(0.0270)	(0.0272)	(0.0215)	(0.0218)	(9034.4)	(9192.9)
BL house-quality controls	No	No	Yes	No	Yes	No	Yes
N	6001	2369	2369	6001	6001	6001	6001

Notes. Quintile 1 = base

◂ Back

First stage: Survey round 1

	Received tenure document		PMAY		
	(1)	(2)	(3)	(4)	(5)
	Self-report	Govt record	Received	Money received	Construction complete
Tenable	0.134***	0.414***	0.0983***	11187.0***	0.0246***
	(0.0144)	(0.0151)	(0.0111)	(1679.2)	(0.00642)
Untenable mean	0.09	0.08	0.04	5480.13	0.01
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5826	5878	5917	5905	5917

Notes. Round-1 survey sample

Back

House quality: LATE of PMAY receipt (round 2)

	House quality index	House quality components				
	(1)	(2)	(3)	(4)	(5)	(6)
		Roof	Wall	Floor	Toilet	Concealed wires
Received PMAY	0.750**	0.801***	0.174	0.228	0.0296	-0.00689
	(0.289)	(0.122)	(0.127)	(0.124)	(0.124)	(0.146)
Untenable mean	-1.7e-09	0.18	0.76	0.76	0.66	0.44
Tenable mean	0.25	0.35	0.82	0.82	0.70	0.51
Slum FE & BL controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5744	6001	6001	6001	6001	6001

Notes. IV estimates of PMAY grant receipt, instrumented by land tenability. Column 1 is the endline house-quality PCA index; columns 2–6 are indicators for a concrete roof, brick wall, covered floor, toilet and concealed wires.

◂ Back

House quality: all outcomes, ITT and LATE (round 2)

Outcome	Untenable mean	ITT		LATE (IV)	
		Coef.	SE	Coef.	SE
House quality index (EL)	-0.000	0.088*	0.035	0.750**	0.274
House quality index (BL weights)	0.571	0.221***	0.040	1.844***	0.327
Concrete roof	0.179	0.096***	0.015	0.801***	0.125
Tiled roof	0.282	0.082***	0.016	0.681***	0.135
Asbestos roof	0.959	0.016*	0.007	0.130*	0.060
Brick walls	0.759	0.021	0.016	0.174	0.118
Mud walls	0.981	0.008	0.006	0.070	0.040
Covered floor	0.764	0.027	0.015	0.228*	0.116
Cement floor	0.764	0.027	0.015	0.228*	0.116
Toilet access	0.658	0.004	0.015	0.030	0.124
In-house water & toilet	0.171	0.003	0.014	0.021	0.117
In-house toilet	0.244	0.004	0.016	0.033	0.129
Exclusive toilet	0.602	0.007	0.016	0.061	0.127
Door with latch	0.907	0.004	0.012	0.030	0.089
Concealed wires	0.445	-0.001	0.018	-0.007	0.145
Number of windows	1.524	0.030	0.056	0.250	0.441
Number of rooms	2.622	-0.098*	0.047	-0.812*	0.403
Satisfaction with house	3.840	0.156**	0.057	1.300**	0.461

◂ Back

How house quality is measured

House quality index: first principal component (PCA) of

- Roof: concrete, tiled, asbestos
- Walls: brick (incl. painted brick), mud
- Floor: tiles, cement
- Toilet: in-house with water, in-house, exclusive, shared
- Safety and space: door with latch, concealed wires, number of windows, number of rooms
- Problems faced in the house (roof leak, waterlogging and others)

Standardised to mean 0 and SD 1 in the untenable group

◂ Back

House quality: alternative index definitions (round 2)

	(1) PCA index (endline)	(2) PCA index (BL weights)	(3) Rent predictors	(4) Satisfaction predictors
Tenable	0.0881*	0.221***	0.0544	0.0821*
	(0.0352)	(0.0402)	(0.0356)	(0.0379)
Untenable mean	-0.00	0.57	-0.00	0.00
Slum FE & BL controls	Yes	Yes	Yes	Yes
N	5744	6001	5744	5744

Notes. PCA index (endline) uses principal-component weights estimated on the endline data. PCA index (BL weights) applies the baseline-survey weights. Column 3 is self-reported satisfaction with the house.

◂ Back

Benefits of a better house: LATE (round 2)

	Satisfaction with house	Rainfall		Outside − inside temp	
	(1)	(2) No roof leak	(3) No waterlogging	(4) All days	(5) Hot days
Received PMAY	1.300**	0.342*	-0.0453	0.642	1.867†
	(0.461)	(0.147)	(0.127)	(0.485)	(1.060)
Untenable mean	3.84	0.23	0.80	1.53	1.59
Slum FE	Yes	Yes	Yes	Yes	Yes
N	5987	5829	5829	5903	1851

Notes. Satisfaction with house is on a 1–5 scale. Outside minus inside temperature is a difference in $^{\circ}$C; hot days are those with outside temperature above 32°C. $p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

◂ Back

Benefits of a better house: Round 1

Panel A: ITT

	Satisfaction with house	Rainfall	
	(1)	(2) No roof leak	(3) No waterlogging
Tenable	-0.00591	0.0234	0.0308*
	(0.0597)	(0.0180)	(0.0141)
Untenable mean	3.91	0.27	0.85
Slum FE	Yes	Yes	Yes
N	5905	5917	5917

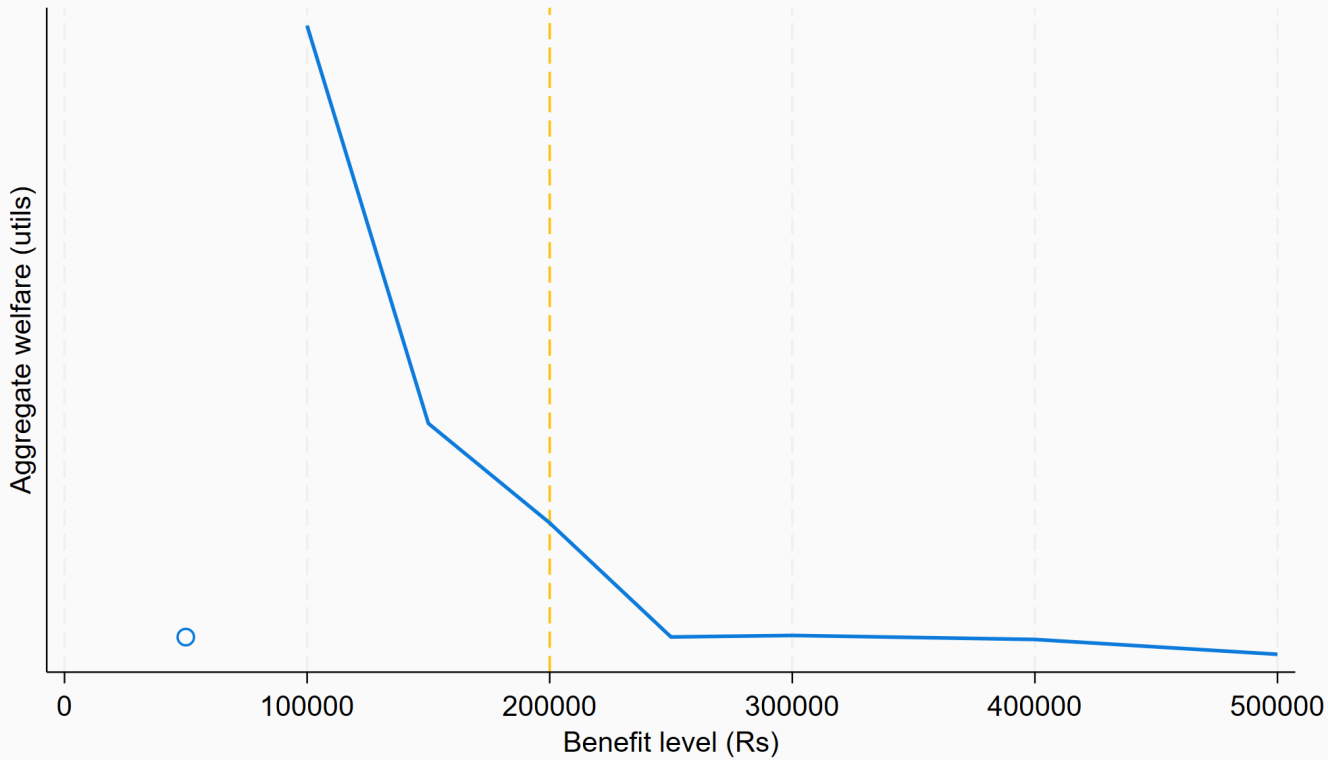
Panel B: LATE (PMAY receipt instrumented by tenability)

	Satisfaction with house	Rainfall	
	(1)	(2) No roof leak	(3) No waterlogging
Received PMAY	-0.0604	0.239	0.314*
	(0.604)	(0.183)	(0.149)
Untenable mean	3.91	0.27	0.85
Slum FE	Yes	Yes	Yes
N	5905	5917	5917

Notes. Round-1 survey sample. Panel A: ITT estimates of tenability. Panel B: IV estimates of PMAY receipt, instrumented by tenability. Satisfaction with house is on a 1–5 scale. $p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

◂ Back

A smaller grant would transfer more welfare



Dashed line: current grant (Rs 200,000)

- Above the current grant, better-housed households select in and the number of recipients falls
- Lower grant: more recipients, and better screening toward those with the most to gain

Back